

Shannon Crandley

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Education

Tufts University

Medford, MA

Bachelor of Science in Computer Engineering

Expected: May 2027

GPA: 3.47/4.00

Relevant Courses: Real-Time Embedded Systems, Digital Signal Processing, Computer Communications Networks, Advanced Computer Architecture, Probabilistic Systems Analysis, Computer Organization, Parallel Computing, Linear Systems, Software Engineering for Regulated Industries, Embedded Systems, Data Structures

Technical Skills

Programming Languages: C, C++, Python, Bash, Java, TypeScript, JavaScript, HTML, CSS, VHDL

Hardware & Protocols: FPGAs, Microcontrollers, ADCs, I²C, SPI, UART, PWM, GPIO, RISC-V, ARM

Software, Frameworks & Tools: Svelte, SvelteKit, Bun, Node.js, Tailwind CSS, PostgreSQL, Drizzle ORM, Docker, Nix, Linux, Git, Jira, MATLAB, LTSpice

Fabrication & Prototyping: Soldering, CAD, 3D Printing, Laser Cutting

Projects

Voice Synthesizer

STM32L432KC Microcontroller, C, ADC, Microphone, LCD

- Developed an embedded C application on an STM32 microcontroller for real-time pitch synthesis
- Sampled microphone input through ADC channels and generated synthesized audio output through a speaker
- Implemented user control via potentiometer and button inputs with LCD display feedback

Self-Navigating Robotic Vehicle

Arduino Uno WiFi Rev2, C, Sensors, Motor Control

- Built an autonomous robot capable of color-based path following and obstacle detection
- Designed sensor systems using phototransistors and IR sensors for color and distance detection
- Implemented motor control with an H-bridge driver and coordinated movement through embedded C firmware
- Developed communication between robots via a server to coordinate movement and stopping behavior

Pac-Man FPGA System

FPGA, VHDL, NES Controller Interface, VGA Display

- Implemented Pac-Man game logic in VHDL on an FPGA, generating real-time graphics through VGA output
- Integrated an NES controller interface for hardware-based user input
- Designed digital modules for player movement, collision detection, and video rendering

Experience

Software Engineer Intern

May 2026 – Aug 2026

Mojotech

Providence, RI

- Engineered a TypeScript application integrating Jira and GitHub APIs to automate Jira issue enhancement with agentic AI, significantly improving project tracking and code modularity
- Participated in code reviews, providing constructive feedback to peers to ensure code quality and adherence to team standards
- Presented the application architecture and impact to the entire company, demonstrating technical communication skills

Research Assistant — MemSysExplorer

Sept 2025 – Dec 2025

Tufts University

Medford, MA

- Developed Python and Bash scripts to automate data collection from CPU and GPU memory benchmarks
- Built analysis pipelines to identify performance trends across different memory architectures
- Generated datasets used for research focused on memory behavior and system performance

Teaching Assistant — Introduction to Computing in Engineering

Tufts University

Jan 2025 – Aug 2025

- Held weekly office hours to assist students with MATLAB programming assignments and debugging
- Graded and provided feedback on assignments for 30+ students

Engineering Outreach Fellow

Tufts University Center for Engineering Education and Outreach

Sept 2024 – Present

- Designed and taught engineering lessons covering robotics, circuits, and CAD
- Mentored middle school students using LEGO Spike Prime robotics platforms